KAIZEN MATHS

Lesson starter

Algebraic Fractions

Multiplying and Dividing

Name _____

S1

Simplify:

$\frac{4(x-4)}{4(x-3)} \times \frac{3(x-3)}{4(x-4)}$

S2

Simplify:

$\frac{3(x-1)}{3x} \div \frac{2(x-1)}{3x}$

S3

Simplify:

$\frac{5(x-5)}{6(x+1)} \times \frac{2(x+1)}{2(x-5)}$

S4

Simplify:

$\frac{5(x+1)}{2(x+3)} \div \frac{2(x+1)}{4(x+3)}$

Show your working

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$\frac{4(x-4)}{4(x-3)} \times \frac{3(x-3)}{4(x-4)}$

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Simplify:

$\frac{3(x-1)}{3x} \div \frac{2(x-1)}{3x}$

S3

Simplify:

$\frac{5(x-5)}{6(x+1)} \times \frac{2(x+1)}{2(x-5)}$

S4

Simplify:

$\frac{5(x+1)}{2(x+3)} \div \frac{2(x+1)}{4(x+3)}$

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Show your working

KAIZEN MATHS

Exit ticket

Algebraic Fractions

Multiplying and Dividing

Name _____

E1

Simplify:

$\frac{5(x-5)}{6(x+2)} \times \frac{2(x+2)}{2(x-5)}$

E2

Simplify:

$\frac{4(x+2)}{5(x-5)} \div \frac{2(x+2)}{3(x-5)}$

E3

Simplify:

$\frac{4(x-5)}{3(x-2)} \times \frac{7(x-2)}{3(x-5)}$

KAIZEN MATHS

Exit ticket

Algebraic Fractions

Multiplying and Dividing

Name _____

E1

Simplify:

$\frac{5(x-5)}{6(x+2)} \times \frac{2(x+2)}{2(x-5)}$

E2

Simplify:

$\frac{4(x+2)}{5(x-5)} \div \frac{2(x+2)}{3(x-5)}$

E3

Simplify:

$\frac{4(x-5)}{3(x-2)} \times \frac{7(x-2)}{3(x-5)}$

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Show your working

KAIZEN MATHS

Exit ticket

Algebraic Fractions

Multiplying and Dividing

Name _____

E1

Simplify:

$\frac{5(x-5)}{6(x+2)} \times \frac{2(x+2)}{2(x-5)}$

E2

Simplify:

$\frac{4(x+2)}{5(x-5)} \div \frac{2(x+2)}{3(x-5)}$

E3

Simplify:

$\frac{4(x-5)}{3(x-2)} \times \frac{7(x-2)}{3(x-5)}$

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Show your working

Teacher answers & working

Algebraic Fractions · Multiplying and Dividing

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S1

Simplify:
$\frac{4(x-4)}{4(x-3)} \times \frac{3(x-3)}{4(x-4)}$

Answer:

$\frac{3}{4}$

Working:

Step 1: Keep the expressions factorised so common factors can be seen.
$\frac{4(x-4)}{4(x-3)} \times \frac{3(x-3)}{4(x-4)}$

Step 2: Cancel the common factor (x - 3) from denominator and numerator.

Step 3: Cancel the common factor (x - 4) from numerator and denominator.

Step 4: Multiply the remaining numerical factors.

$\frac{4\times3}{4\times4} = \frac{3}{4}$.

Step 5: The simplified answer is $\frac{3}{4}$.

S2

Simplify:
$\frac{3(x-1)}{3x} \div \frac{2(x-1)}{3x}$

Answer:

$\frac{3}{2}$

Working:

Step 1: Dividing by a fraction means multiplying by its reciprocal.
$\frac{3(x-1)}{3x} \div \frac{2(x-1)}{3x} = \frac{3(x-1)}{3x} \times \frac{3x}{2(x-1)}$

Step 2: Cancel the common factor x.

Step 3: Cancel the common factor (x - 1).

Step 4: Multiply the remaining numerical factors.

$\frac{3\times3}{3\times2} = \frac{3}{2}$.

Step 5: The simplified answer is $\frac{3}{2}$.

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S3

Simplify:
$\frac{5(x-5)}{6(x+1)} \times \frac{2(x+1)}{2(x-5)}$

Answer:

$\frac{5}{6}$

Working:

Step 1: Keep the expressions factorised so common factors can be seen.
$\frac{5(x-5)}{6(x+1)} \times \frac{2(x+1)}{2(x-5)}$

Step 2: Cancel the common factor (x + 1) from denominator and numerator.

Step 3: Cancel the common factor (x - 5) from numerator and denominator.

Step 4: Multiply the remaining numerical factors.

$\frac{5\times2}{6\times2} = \frac{5}{6}$.

Step 5: The simplified answer is $\frac{5}{6}$.

S4

Simplify:
$\frac{5(x+1)}{2(x+3)} \div \frac{2(x+1)}{4(x+3)}$

Answer:

5

Working:

Step 1: Dividing by a fraction means multiplying by its reciprocal.
$\frac{5(x+1)}{2(x+3)} \div \frac{2(x+1)}{4(x+3)} = \frac{5(x+1)}{2(x+3)} \times \frac{4(x+3)}{2(x+1)}$

Step 2: Cancel the common factor (x + 3).

Step 3: Cancel the common factor (x + 1).

Step 4: Multiply the remaining numerical factors.

$\frac{5\times4}{2\times2} = 5$.

Step 5: The simplified answer is 5.

Teacher answers & working

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E1

Simplify:
$\frac{5(x-5)}{6(x+2)} \times \frac{2(x+2)}{2(x-5)}$

Answer:

$\frac{5}{6}$

Working:

Step 1: Keep the expressions factorised so common factors can be seen.
$\frac{5(x-5)}{6(x+2)} \times \frac{2(x+2)}{2(x-5)}$

Step 2: Cancel the common factor (x + 2) from denominator and numerator.

Step 3: Cancel the common factor (x - 5) from numerator and denominator.

Step 4: Multiply the remaining numerical factors.

$\frac{5\times2}{6\times2} = \frac{5}{6}$.

Step 5: The simplified answer is $\frac{5}{6}$.

E2

Simplify:
$\frac{4(x+2)}{5(x-5)} \div \frac{2(x+2)}{3(x-5)}$

Answer:

$\frac{6}{5}$

Working:

Step 1: Dividing by a fraction means multiplying by its reciprocal.
$\frac{4(x+2)}{5(x-5)} \div \frac{2(x+2)}{3(x-5)} = \frac{4(x+2)}{5(x-5)} \times \frac{3(x-5)}{2(x+2)}$

Step 2: Cancel the common factor (x - 5).

Step 3: Cancel the common factor (x + 2).

Step 4: Multiply the remaining numerical factors.

$\frac{4\times3}{5\times2} = \frac{6}{5}$.

Step 5: The simplified answer is $\frac{6}{5}$.

Teacher answers & working

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E3

Simplify:
$\frac{4(x-5)}{3(x-2)} \times \frac{7(x-2)}{3(x-5)}$

Answer:

$\frac{28}{9}$

Working:

Step 1: Keep the expressions factorised so common factors can be seen.
$\frac{4(x-5)}{3(x-2)} \times \frac{7(x-2)}{3(x-5)}$

Step 2: Cancel the common factor (x - 2) from denominator and numerator.

Step 3: Cancel the common factor (x - 5) from numerator and denominator.

Step 4: Multiply the remaining numerical factors.

$\frac{4\times7}{3\times3} = \frac{28}{9}$.

Step 5: The simplified answer is $\frac{28}{9}$.